

# An incident copilot that shows its work.

Connect voice-led investigation to cited observations, explicit approvals, and recovery verification.

- Built for the AssemblyAI Voice Agent Hackathon
- Voice triage, explicit approvals, evidence-based incident reviews

# Keep the investigation moving.

During an outage, engineers move between conversations, logs, service health, and follow-up notes.

- A successful command is not proof of a recovered service.
- Engineers need to inspect the diagnosis, control the action, and verify its effect.
- Target users: on-call engineers and small platform teams.

# From a question to a reviewed action.

Ask about the incident, inspect evidence, and decide what should happen next.

- Investigate → capture health and logs, then inspect evidence-linked hypotheses.
- Stage → show the exact action, service, parameters, and approval expiry.
- Approve or cancel → execute once, then show the observed result.
- Verify → compare before/after health and check the remaining incident impact.

# Two voice paths. One tool boundary.

Both voice engines share the same session and approval logic.

- Managed: AssemblyAI Voice Agent API with 24 kHz PCM audio and client tools.
- Custom: AssemblyAI Streaming v3 STT → configured LLM or labeled scripted commands → speech output.
- Investigation briefs and reports: AssemblyAI LLM Gateway analyzes captured evidence.
- Provider failures and local fallbacks are visible in the workspace.

# Approval is part of execution.

A model request stages a mutation. A second tool call does not authorize it.

- Each action has a unique ID, exact target, and a 30-second approval window.
- Cancellation, expiry, role checks, and replay rejection are enforced in the backend.
- Live infrastructure requires an operator token.
- Automatic approval is restricted to explicitly enabled simulation.

# Open the observation behind the hypothesis.

**IncidentVoice**  
VOICE INCIDENT WORKSPACE

STANDBY STT: - Total: - Settings

OPERATIONS / INCIDENT WORKSPACE

**Less typing. Faster triage.**  
Investigate together. Make the next move with confidence.

Investigate Incident Create Incident report

SEV-1 INC-8942

**Payment Gateway Outage (HTTP 503 Spike)**

Investigating / Demo simulation · no live changes

NEEDS ATTENTION  
4  
2 active critical services

CLUSTER HEALTH  
1 / 5  
Healthy in this scenario

**Voice SRE Incident Copilot**  
Speak naturally. Stay in control. Copilot Standby

Operator 02:08:51 PM

Investigate the incident

IncidentVoice SRE 02:08:54 PM

4 services need investigation. 17 observations captured; root cause remains unverified. AssemblyAI proposed 3 hypotheses to test. Review the cited evidence before approving a change.

QUICK Investigate the incident What alerts are firing? Inspect payment-service Show dependency topology

Start voice MIC STANDBY

Type SRE command or speak into mic...

Enter to send Changes require your approval

INVESTIGATE  
The bigger picture 5 services

Brief Services Runbooks Topology Demo lab

Investigate → Review & approve → Verify

simulation evidence Captured 02:08 PM Refresh

ASSEMBLYAI INVESTIGATION

**What the evidence suggests**

4 services need investigation. 17 observations captured; root cause remains unverified. AssemblyAI proposed 3 hypotheses to test.

17 observations · 5 services

01 HYPOTHESIS · UNVERIFIED  
**Database connection exhaustion and lock contention**

The database has hit its maximum connection limit, causing the payment-service to exhaust its connection pool and crash due to memory pressure while waiting for locks.

Supporting observations E01 ↓ E02 ↓ E03 ↓ E04 ↓ E09 ↓

**Next check**  
Inspect the latest logs for order-db and compare them with the cited snapshot before choosing a

Captured facts.  
Inspectable hypotheses.

- 17 observations from five simulated services in this demonstration.
- AssemblyAI analysis linked to exact evidence IDs.
- Read-only next checks; stale evidence is flagged.

# A healthy target. An incident still in progress.

The screenshot displays the IncidentVoice 'VOICE INCIDENT WORKSPACE'. At the top, it shows 'STANDBY' status and 'Settings'. The main header reads 'Less typing. Faster triage.' with a subtext 'Investigate together. Make the next move with confidence.' and buttons for 'Investigate Incident' and 'Create Incident report'.

The incident details section shows 'SEV-1 INC-8942' titled 'Payment Gateway Outage (HTTP 503 Spike)'. It is marked as 'Investigating' with a note 'Demo simulation - no live changes'. Summary cards indicate '3' items 'NEEDS ATTENTION' (1 active critical services) and '2 / 5' 'CLUSTER HEALTH' (Healthy in this scenario).

The 'Voice SRE Incident Copilot' section on the left shows a chat log:
 

- IncidentVoice SRE (02:08:55 PM): Simulation applied. Graceful rolling restart executed for payment-service. Clean pods spawned.
- Operator (02:08:55 PM): Verify recovery
- IncidentVoice SRE (02:08:55 PM): Recovery is incomplete: ingress-gateway, order-db, redis-cache still need attention or health verification.

 Below the chat is a 'Start voice' button and a text input field for SRE commands.

The 'INVESTIGATE' section on the right, titled 'The bigger picture', shows a 'Brief' tab with a 'simulation' button. It displays 'AFTER THE ACTION' results:
 

- The target is healthy.**
- payment-service - restart pod
- Status: critical → healthy
- Error rate: 42.6% → 0.5%
- P99 latency: 2850 ms → 65 ms

 A note states: 'Simulated result. This check covers the target service; other services may still be affected.' A 'Check the whole cluster' button is present.

At the bottom of the 'INVESTIGATE' section, a status box says '✓ Recovery is still in progress' with the note 'Recovery is incomplete: ingress-gateway, order-db, redis-cache still need attention or health verification.' Below this is a 'simulation evidence' box with a timestamp 'Captured 02:08 PM' and a 'Refresh' button. A final warning states: 'Service state has changed since this brief. Refresh before relying on its hypotheses.'

Success needs context.

- Approve the exact service restart.
- Compare observed before/after target health.
- Check the whole cluster: three services still need attention here.
- Export the evidence and outcomes for the next engineer.

# A focused prototype with a clear path.

The next milestone is a repeatable, measured pilot with an on-call team.

- Current limits: in-memory single-process sessions and limited live infrastructure actions.
- Validate microphone quality, interruptions, and response latency on the deployment target.
- Add persistent evidence storage, individual identities, and deeper telemetry integrations.
- Supply the final repository, HTTPS app, and demo video links in the submission form.